When Does Adaptive Recovery Pay? Crossover-Horizon Prediction for Failure Recovery under a Noisy Failure Classifier

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***Abstract*—Tool-using agents fail in recurring and classifiable ways, and the choice of recovery strategy is increasingly treated as something to be learned rather than fixed in advance. A designer weighing a hand-written recovery table against an online learner needs to know at what deployment horizon the learner overtakes the table. Existing crossover analyses assume the policy observes the true failure type, which real systems do not provide; the policy sees a classifier's prediction. We reformulate the crossover problem under an arbitrary confusion matrix by mapping the strategy-success matrix into predicted-type space with an exact Bayes mixture, then separate the resulting degradation into a reward-ceiling channel and a learnability channel. At a classifier with 0.939 mean diagonal accuracy, ceiling loss exceeds learnability loss by roughly six to one. The two channels also work against each other: classifier noise raises the static table's regret rate by 58% yet shifts the predicted crossover only from 3053 to 3126 episodes, and we show by construction that the horizon is not monotonic in classifier accuracy. The reformulation's free constant was tested under a protocol registered before the supporting data existed. The held-out ratio $R = 1.4729$ clears its pre-committed threshold of 1.5, but the bootstrap interval $[1.00, 9.55]$ required by the same protocol does not confirm that the clearance is stable, so we report the component as partially supported.**

***Index Terms*—autonomous agents, failure recovery, multi-armed bandits, Thompson sampling, noisy context, pre-registration, negative results**

I. Introduction

Language agents that call external tools fail often enough that failure handling has become a design concern in its own right. Timeouts, tool exceptions, malformed outputs and lost context dominate the failure inventories of both deployed and benchmarked systems [1]–[3]. The usual answer is a fixed mapping π from failure type to recovery strategy. Such tables are cheap and easy to inspect, and can be quite good, but they never improve; a table is exactly as good as its author's beliefs on the day it was written.

The alternative is to treat each pair of failure type and recovery strategy as a bandit arm and learn its value from outcomes. Adaptation then costs exploration: early in a deployment the learner does worse than a good table because it is still paying to find out what works, so whether adaptation is worthwhile depends on how long the system will run. The designer wants a number, the horizon at which the two curves cross.

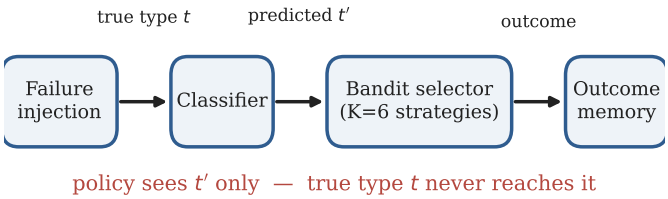


Fig. 1. Where the classifier sits. Policies observe only the predicted type t' and never the true type t, which is the structural fact motivating Section IV.

A crossover predictor supplies that number. It computes the table's constant per-episode regret rate exactly, models the learner's cumulative regret with an asymptotic bound in the tradition of Lai and Robbins [4], and solves for the point where the two meet. The construction assumes the policy observes the true failure type, and deployed systems do not supply one. Tracing the call chain of the framework studied here confirmed that no policy can read a true type; the context object has no such field, so every policy acts on a classifier's prediction (Fig. 1). This is a property of the system rather than a modelling choice, and it means the unmodified predictor answers a question about a system nobody has.

This paper works through the resulting extension. We give an exact reformulation of the crossover problem under an arbitrary confusion matrix, using a Bayes mixture that carries the success matrix into predicted-type space and reduces to the original construction when the classifier is perfect. We decompose classifier-induced degradation into ceiling loss and gap shrinkage, measure both at a trained classifier's empirical matrix, and show that the predicted horizon is not monotonic in accuracy. We then evaluate the extension's one free constant on a 26-environment dataset generated after the evaluation protocol was committed. A companion study, pre-registered in the same way, asks whether the sign of adaptive advantage under drift reduces to a low-dimensional statistic; it does not,

and we report that as a negative result. What we do not claim is that the extension makes the underlying asymptotic bound accurate: held-out absolute errors stay near 40–60%, and the evaluation asks only whether the reformulation adds error on top of what the base bound already carries.

II. Related Work

Work on agent failure has moved from anecdote toward structured taxonomy, whether organised by multi-agent failure mode [1], by single-agent module with a debugger that isolates the earliest outcome-changing error [2], or by the complementary route of improving the environment rather than the agent [3]. On the response side, Reflexion writes reflections on failed attempts into an episodic buffer [5], CRITIC calls external tools to verify and revise the model's own output [6], and PALADIN trains recovery behaviour from systematically injected tool failures [7].

Our concern is different. We propose no new self-correction mechanism, and ask instead how a recovery-strategy selector should be evaluated against a static baseline once selection is treated as a bandit problem with persistent outcome memory. The underlying trade-off is classical: Lai and Robbins established the logarithmic lower bound [4], Auer et al. showed logarithmic regret is achievable uniformly in time [8], and Thompson sampling [9], whose crossover behaviour we study, has a modern treatment in Russo et al. [10].

The specific difficulty here, a learner conditioning on a corrupted view of the context, has an established literature. Kirschner and Krause study bandits in which the environment supplies a context distribution while the realisation stays hidden [11]; Kim et al. find that under noisy features the Bayes-optimal hypothesis can sit far from the underlying realizability function [12]; and latent bandits, where arm rewards are known conditional on an unobserved discrete state, are the closest structural analogue to a failure type the policy never sees [13], [14]. Our mixture is a finite, discrete specialisation of the same marginalisation idea and we claim no novelty for it. Three things are ours: the corruption channel is a measured confusion matrix rather than an assumed noise model; the quantity of interest is a crossover horizon against a hand-written baseline, which is what a practitioner acts on, not a regret rate; and we treat the free multiplicative constant as an object to be calibrated and held-out tested rather than folded into $O(\cdot)$.

III. Setup

A. Environments and policies

Failures come from $N = 7$ types and are recovered with one of $K = 6$ strategies. The types are `tool_timeout`, `tool_error`, `wrong_tool_selected`, `hallucinated_output`, `malformed_output`, `context_loss` and `ambiguous_instruction`; the strategies are `retry`, `replan`, `switch_tool`, `re_ground`, `verify` and `clarify`.

An environment is a pair (P, π). The matrix $P \in [0,1]^{N\times K}$ gives the success probability $P[t,k]$ of strategy k against true type t, with conditionally Bernoulli rewards. The mapping π is a hand-authored table, and p is the distribution over true types. Failures are injected into simulated episodes, classified, and recovered by a bandit selector whose outcomes persist across episodes.

One environment, which we call canonical, has a hand-authored P and a hand-authored π agreeing with the row argmax of P on six of seven types; we call this quality 6/7, a competent baseline rather than a straw man. The other 25 environments are sampled. Each row draws a uniform baseline in $[0.15, 0.55]$, then one randomly chosen strategy in that row is raised to the row maximum plus a fixed gap, capped at 0.95; all results here use gap $= 0.20$. Those 25 reuse the canonical table instead of receiving one authored against their own P, so baseline quality across the sweep is uncontrolled, and in two environments the reused table matches none of the seven row argmaxes; the 6/7 figure describes the canonical environment only. Section VI-C returns to this.

B. The predictor being extended

We restate the base construction so the paper stands on its own. Write

$$\mathrm{oracle}(P,p) = \sum_t p[t] \max_k P[t,k], \tag{1}$$

where $t \in \{1,\dots,N\}$ indexes the true failure type, $k \in \{1,\dots,K\}$ indexes the recovery strategy, $p[t]$ is the probability that an episode's failure is of type t, and $P[t,k]$ is the probability that strategy k recovers a type-t failure; $\mathrm{oracle}(P,p)$ is thus the per-episode success rate of a policy that always plays the best strategy for the true type. Correspondingly,

$$\mathrm{policy}(P,\pi,p) = \sum_t p[t]\, P[t,\pi(t)], \tag{2}$$

where $\pi(t) \in \{1,\dots,K\}$ is the strategy the hand-written table assigns to type t, so $\mathrm{policy}(P,\pi,p)$ is the table's per-episode success rate.

The table's per-episode regret rate $\rho = \mathrm{oracle}(P,p) - \mathrm{policy}(P,\pi,p)$ is constant, so its cumulative regret is exactly ρT. The learner's cumulative regret is modelled by a Lai–Robbins-style bound in KL form,

$$R_{\mathrm{TS}}(T) = c\,N \sum_t p[t] \sum_{k\,:\,P[t,k]<P[t,k_t^*]} \frac{\log\big(\max(T/N,1)\big)}{\mathrm{KL}\big(P[t,k],\,P[t,k_t^*]\big)}, \tag{3}$$

where T is the deployment horizon in episodes, $k_t^* = \arg\max_k P[t,k]$ is the best strategy for type t, the inner sum runs over the strictly suboptimal strategies in row t, $\mathrm{KL}(a,b)$ is the Bernoulli divergence clipped away from 0 and 1 for numerical stability, N is the number of types, and $c > 0$ is a free multiplicative constant. The crossover horizon T^* solves $R_{\mathrm{TS}}(T) = \rho T$. The constant c has no closed-form value for a given problem instance and must be fitted; under confusion we write it κ. It is the single free parameter whose transferability we test in Section VI-C, and everything downstream of it is calibration rather than proof.

Let $\mathrm{confusion}[t,t'] = \Pr(\text{predicted } t' \mid \text{true } t)$. The base predictor implicitly sets this to the identity matrix I. Two

questions follow. First, what is the right crossover prediction once the policy conditions on t'? This turns out to have an exact answer on the static side and only an approximate one on the adaptive side. Second, does a constant fitted under one classifier condition transfer to another? If a κ fitted on perfect-classifier data predicts noisy held-out cells about as well as one fitted directly on noisy data, the reformulation has absorbed the classifier effect and the remaining error is inherited. If not, the reformulation is adding error of its own. Separating these requires held-out data and a decision rule fixed before looking.

IV. Crossover Under Classifier Noise

A. The effective mixture

Marginalising over true types gives the predicted-type distribution

$$p'[t'] = \sum_t p[t]\,\mathrm{confusion}[t, t'], \tag{4}$$

where t' indexes the type the classifier reports, $\mathrm{confusion}[t, t']$ is the probability that a true type t is reported as t', and $p'[t']$ is therefore the probability that an episode lands in predicted bucket t'. Bayes' rule inside each bucket gives the effective success probability of strategy k given a prediction of t':

$$P_{\text{eff}}[t', k] = \frac{\sum_t p[t]\,\mathrm{confusion}[t, t']\,P[t, k]}{p'[t']}, \tag{5}$$

where the numerator sums the contribution of every true type that can be reported as t', weighted by how often that happens, and the denominator normalises by the bucket's total mass, so $P_{\text{eff}}[t', k]$ is the probability that strategy k succeeds when the classifier says t'.

Both are exact and both collapse to $p' = p$, $P_{\text{eff}} = P$ at $\mathrm{confusion} = I$. Equation (5) is the object the rest of the extension is built on: given that the classifier said t', how well does strategy k really work, averaged over what the type actually was.

B. Static and adaptive sides

The static side composes two exact functions,

$$\rho_{\text{conf}} = \mathrm{oracle}(P, p) - \sum_t \sum_{t'} p[t]\,\mathrm{confusion}[t, t']\,P[t, \pi(t')], \tag{6}$$

where $\pi(t')$ is the strategy the table fires on seeing the *predicted* type t', while $P[t, \pi(t')]$ scores that strategy against the *true* type t; ρ_{conf} is thus the table's per-episode regret rate once its input is corrupted, and it is exact rather than asymptotic. The adaptive side reuses (3) unmodified, evaluated on the mixture,

$$R_{\text{TS}}^{\text{conf}}(T) = R_{\text{TS}}(P_{\text{eff}}, T, N, p', \kappa), \tag{7}$$

where P_{eff} and p' replace P and p from (3), N is unchanged, and κ is the fitted constant playing the role of c. The horizon under confusion solves $R_{\text{TS}}^{\text{conf}}(T) = \rho_{\text{conf}} T$.

This is its own routine rather than a wrapper on the base solver, because it mixes index spaces: ρ_{conf} lives in true-type space while the regret term lives in predicted-type space. Unit tests confirm that at $\mathrm{confusion} = I$ the extension reproduces the base solver bit for bit, so it is a strict generalisation rather than a parallel implementation that happens to agree.

C. Two channels, two checks

Classifier noise does damage in two ways that need not move together. Merging true types into indistinguishable buckets destroys information and lowers the achievable ceiling whatever the policy does,

$$\Delta_{\text{ceil}} = \mathrm{oracle}(P, p) - \mathrm{oracle}(P_{\text{eff}}, p'), \tag{8}$$

where the first term is the best achievable per-episode success rate with perfect type information and the second is the best achievable rate once types have been merged into predicted buckets, so $\Delta_{\text{ceil}} \geq 0$ measures reward permanently lost to merging, with equality exactly at $\mathrm{confusion} = I$. Merging also changes the gap between the best and second-best action inside each bucket, which changes how hard the arm is to identify. We report the mean shrinkage of that top-two gap across buckets, computed as an unweighted mean over contributing true rows. The first is a ceiling effect, the second a learnability effect.

Two checks guard the construction. The first asks whether a bucket's best action under P_{eff} could be optimal for none of the true types feeding it; it does not fire at our measured confusion, and a constructed three-type example confirms it does fire when its structural condition holds. The second concerns monotonicity: while ρ_{conf} and Δ_{ceil} are provably monotonic in classifier accuracy, T^* is not. A two-type, two-strategy unit test has a bucket whose best arm changes identity as accuracy crosses a threshold, making its top-two gap V-shaped and zero at the flip, whatever κ is.

D. Assumptions

Five assumptions carry the extension and they are not equally safe. A1, the predicted-type bucket structure, was verified by tracing the implementation; A2, conditionally Bernoulli rewards, is inherited unchanged. A3, a stationary confusion matrix, is optimistic wherever the classifier itself degrades. A4, the asymptotic bound, holds as $T \to \infty$ rather than at finite time and still carries an uncalibrated constant. A5, scalar accuracy standing in for a full matrix, is a simplification we sidestep by validating against the classifier's actual, very non-uniform matrix.

V. Protocol

The classifier combines a TF-IDF representation (word 1–3 grams, 5000 features) with a one-hot tool identifier, feeding a linear SVM with balanced class weights; calibrated probabilities come from a separate logistic regression. It is trained on 300 template-generated failure events from the same synthetic injector the simulator uses, and the matrix in Fig. 2 was measured on a disjoint batch of 150 events per type under a different seed. The 0.939 accuracy is held out with respect to instances but not with respect to distribution, so the matrix should be read as a plausible *shape* of classifier error,

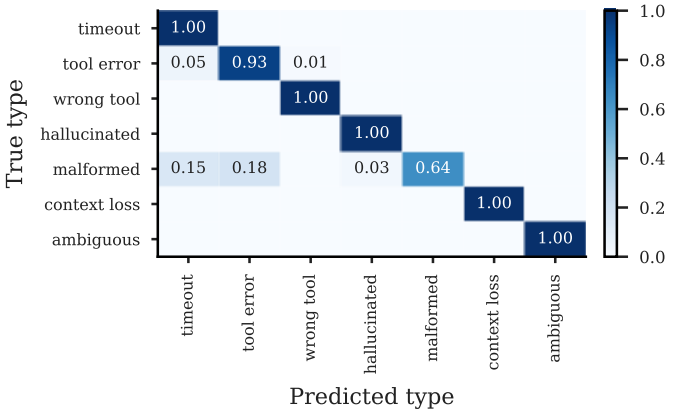


Fig. 2. Empirical confusion matrix of the trained classifier, Pr(predicted | true), mean diagonal accuracy 0.939. Five of seven rows are effectively identity rows.

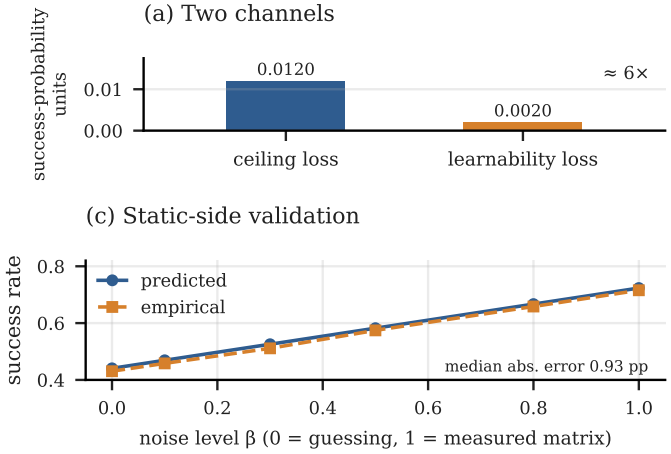


Fig. 3. Decomposition at the measured classifier and static-side validation. (a) Ceiling loss is about six times larger than learnability loss. (b) The error is concentrated: one row carries roughly 0.36 of the off-diagonal mass while five rows are identity rows. (c) The exact static form of (6) tracks simulation across six noise levels β, 50 seeds each, median absolute error 0.93 percentage points.

concentrated and asymmetric with one dominant confusable type, and not as an accuracy that would survive production failure text.

The evaluation protocol was written and committed before any of the supporting data existed. Of the 25 gap-family environments, 8 went to TEST by a fixed-seed draw and the one non-gap-family environment stayed in TRAIN unconditionally, giving 18 TRAIN and 8 TEST; two environments already known to have poor errors landed in TEST and we did not redraw. Fitting used bounded 1-D least squares on mean squared log-space error with the KL form fixed. Branch A (transfer) fits κ_A on TRAIN perfect-classifier cells and applies it unchanged to TEST noisy cells; Branch B (in-condition) fits κ_B on TRAIN noisy cells and applies it to the same TEST cells, giving the best achievable same-condition benchmark. The decision rule was $R = E_A/E_B$ on median absolute relative error, with $R \leq 1.5$ committed to Case 1, the effect is absorbed and residual error inherited, and $R > 1.5$ to Case 2, the extension adds error of its own, reported with no rescue attempt. Uncertainty came from an environment-level bootstrap with 2000 resamples, and a constant landing within one order of magnitude of the base predictor's own best fit, $[0.00045, 0.045]$, was to be called a calibrated predictor rather than a validated bound. One specification, no second attempt.

VI. Results

A. Decomposition and static-side validation

At the classifier's empirical matrix (Fig. 2) the ceiling reduction is $\Delta_{\text{ceil}} = 0.0120$ and the mean gap shrinkage across the seven buckets is 0.0020, so ceiling loss runs about six times larger than learnability loss (Fig. 3a). Both are real and neither is zero, but the bulk of the measured degradation is information lost to merging rather than a harder arm-identification problem. The split is specific to this confusion structure and should not be read as a general ratio. The concentration is severe (Fig. 3b): `malformed_output` alone carries roughly 0.36 of the off-diagonal mass and `tool_error` a further 0.067, while the remaining five rows are effectively identity rows, which is why the two losses are not spread evenly across buckets. The argmax-inversion check fires in none of the seven.

We then checked the exact static component of (6) against an existing table of six noise levels with 50 simulated seeds each, where β interpolates linearly between uniform-random type prediction and the measured matrix, so $\beta = 0$ is a classifier that guesses and $\beta = 1$ is the real one. That table carries no adaptive trajectory, so it validates the static side alone and is never used as evidence for the adaptive theory. As Fig. 3c shows, the median absolute error is 0.93 percentage points and no level exceeds 1.5. This is the strongest validation result anywhere in this work, which is about what one expects from closed-form arithmetic rather than an asymptotic approximation.

B. The horizon itself

Applying the fitted transfer constant to the canonical environment produces the number the construction exists to give (Fig. 4). Under a perfect classifier the predicted crossover falls at 3053 episodes; under the measured matrix it falls at 3126.

The direction repays a moment's attention, because the two channels pull against each other and the smaller one wins. Classifier noise raises the table's per-episode regret rate from $\rho = 0.0214$ to $\rho_{\text{conf}} = 0.0338$, a 58% increase from a classifier that is right 94% of the time, and that alone would pull the crossover earlier: a baseline bleeding faster is caught sooner. It does not, because the learner's problem also became harder. Ceiling reduction and gap shrinkage enter (3) through inverse-KL denominators sensitive to small changes in the top-two gap, and the net effect moves the horizon about 2.4% later. This near-cancellation belongs to this environment and this matrix; Fig. 5 sweeps the same effect over accuracy in a minimal two-type instance, where the horizon diverges where the bucket's best arm changes identity. Given held-out errors

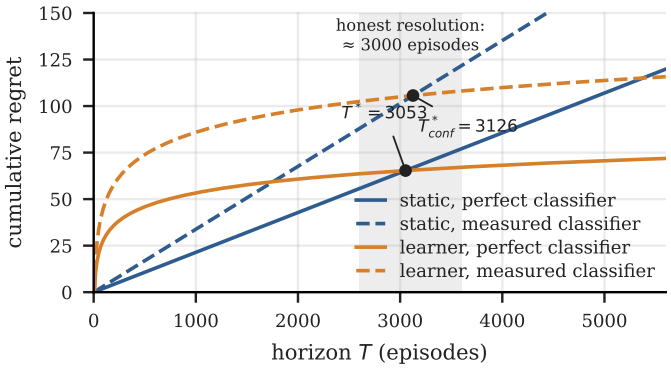


Fig. 4. The crossover on the canonical environment. Classifier noise steepens the static baseline by 58% but also flattens the learner's curve, so the horizon moves 2.4% later rather than earlier.

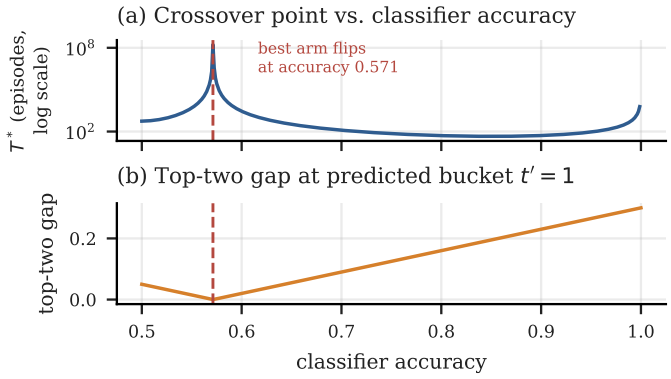


Fig. 5. A better classifier does not imply an earlier crossover. T^* diverges at the accuracy where the bucket's best arm flips and its top-two gap vanishes.

of 40–60%, the honest resolution of these figures is "roughly three thousand episodes", not 3053 against 3126.

C. Held-out evaluation of the constant

The power check cleared comfortably, with 18 of 18 TRAIN and 8 of 8 TEST cells finite in both branches against a floor of 5. Fig. 6 has the headline numbers: Branch A reaches 59.7% median absolute relative error against Branch B's 40.5%, for $R = 0.5970/0.4053 = 1.4729$.

The point estimate satisfies the committed rule, making this a Case 1 outcome: the mixture captures the classifier-noise effect and what error remains is inherited asymptotic-bound bias rather than something the extension introduced. Two things temper that reading. Absolute errors of 59.7% and 40.5% are large, so the test shows a pre-existing limitation transferring cleanly rather than being resolved. And the margin is 1.8%, with the pre-registered bootstrap interval for R running to 9.55, deep into Case 2, so a different draw of held-out environments could plausibly have landed there; the fit-uncertainty bootstrap agrees that the noisy-condition fit is markedly less stable than the transferred one (Fig. 6a).

Fig. 6b shows why the medians should not be read as a tight summary: two environments over-predict by roughly 5.75× under both constants and one under-predicts by roughly 3×.

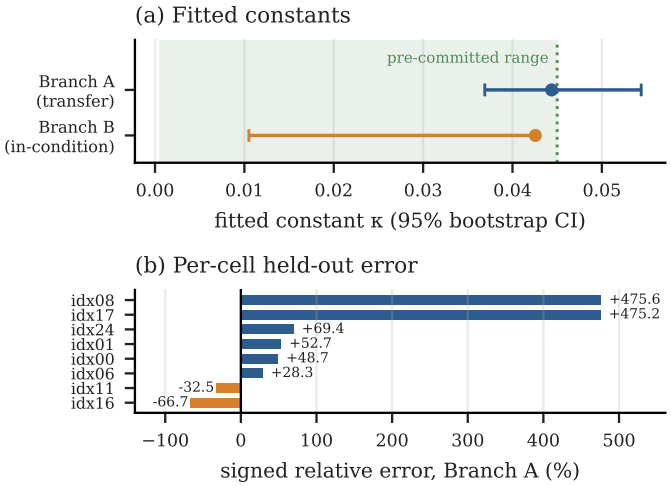


Fig. 6. Pre-registered held-out evaluation on the eight TEST cells. (a) Fitted constants with 95% bootstrap intervals; both sit inside the pre-committed range and the in-condition fit is the less stable of the two. (b) Branch A per-cell signed relative error, a spread no scalar constant can absorb.

No scalar constant fixes this spread, and the protocol's no-rescue commitment meant we did not try to introduce one. One candidate explanation deserves naming. The 25 sampled environments reuse the canonical table, so baseline quality varies uncontrolled and is very low in some cells; baseline quality sets ρ, and ρ sets the slope the regret curve must cross, so the heterogeneity plausibly tracks baseline quality rather than anything about the reformulation. Controlling it, or entering it as a pre-registered covariate, is the highest-priority fix to this design. We flag it rather than analyse it now, because running a quality covariate after seeing the per-cell errors is exactly the exploratory move the protocol exists to prevent.

The fitted transfer constant, 0.0444, sits inside the pre-committed calibrated range near its upper edge. We therefore call this component a calibrated predictor with theoretical motivation rather than a validated bound, and classify the crossover extension overall as partially supported. Case 1 was the pre-committed outcome and is real held-out evidence for the first time in this line of work, but the fragility is equally real. Equations (4)–(5), (6) and (8), the gap-shrinkage report, the argmax-inversion check and the non-monotonicity finding are all exact and validated; only (7) and the horizon inherit the asymptotic approximation. All of it is simulated, including the classifier, so the numbers describe the construction rather than any live agent.

D. Drift: a negative result

A separate extension, independent of the first, asks whether a low-dimensional statistic can predict the sign of adaptive advantage under environment drift, out of sample. We fixed the falsifiable structure first: a candidate must beat a majority-class guess under leave-one-cell-out evaluation. Three candidates were pre-registered and tried in order over a pre-simulated grid of 12,800 cells across four drift mechanisms, with a fourth ruled out in advance if the third failed. Fig. 7a summarises the

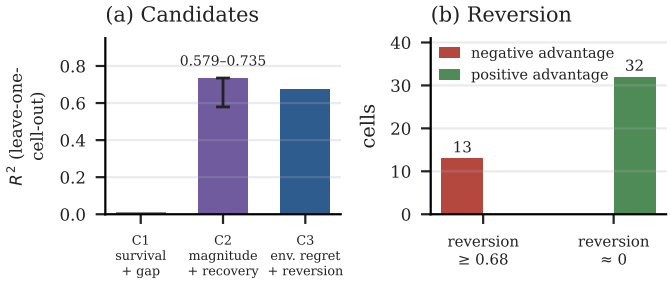


Fig. 7. The drift study. (a) Three pre-registered candidate statistics: C1 is insufficient, C2 was retracted for circularity, C3 fits well but its sign accuracy only equals the majority-class baseline. (b) The one qualitative finding that survives: adaptive advantage is negative exactly where staleness damage reverts and positive where it does not.

outcome. Survival and gap-change was insufficient. Magnitude and recovery-ratio fitted well but was retracted before it could count as a finding, because its recovery-ratio term has the base predictor in its own denominator, so testing whether it predicts adaptive advantage was largely testing whether the base bound has some validity applied row by row. Environment regret and reversion reached $R^2 = 0.676$ but its sign accuracy exactly equalled the majority baseline it had to beat. One methodological wrinkle is worth recording: first-attempt statistics were means over touched rows only, and most of magnitude's apparent contribution in the second attempt ($+0.156$ R^2) was an artifact of that exclusion, shrinking to $+0.006$ when recomputed at the environment level.

One qualitative finding survives the failed quantitative rules: the sign of adaptive advantage under drift is governed by whether staleness damage reverts, not by how large it is (Fig. 7b). All 13 negative-advantage cells sit at a reversion fraction of at least 0.68 across 260 seed draws, and all 32 cells with reversion near zero come out positive across 640 draws. This also explains why the static table beats the learner in 10 of 16 partial-reorder cells: best and second-best swap on a period-2 cycle there, so the table's stale action stays near-optimal while the learner pays exploration cost chasing a target that returns to where it began. We treat this as correlational with a mechanism behind it, and drift-type fixed effects retain $+0.092$ R^2 beyond the two-statistic summary, structure we have not accounted for.

VII. Conclusion

We extended a crossover-horizon predictor to the setting real systems occupy, where recovery policies see a classifier's prediction rather than the true failure type. The extension is exact where it can be, collapses to its predecessor at a perfect classifier, and separates classifier-induced degradation into a ceiling channel and a learnability channel, the former about six times the latter at our measured accuracy. On the canonical environment the two channels nearly cancel, leaving the horizon almost unmoved despite a 58% rise in the baseline's regret rate, and the horizon is not monotonic in accuracy, which rules out an intuitive design heuristic. The free constant was evaluated under a protocol fixed before the data existed: the criterion is satisfied by point estimate but its uncertainty quantification does not confirm that the satisfaction is stable, so we call the component a calibrated predictor rather than a validated bound. The companion drift study stands as a negative result with one qualitative finding intact.

Three directions follow, in order of priority. A live-agent evaluation should measure P and the confusion matrix on real tool failures, since every number here rests on a synthetic template generator and an in-distribution classifier accuracy that real failure logs would not be expected to reproduce. Static baseline quality should then be controlled across the environment family, or entered as a pre-registered covariate, so that the per-cell spread in Fig. 6b can be attributed rather than merely flagged. Finally, relaxing the stationary-confusion assumption would let the two extensions connect, and a causal drift test with damage magnitude matched and reversion varied alone would turn the surviving qualitative finding into a tested mechanism.

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